Tejas Salian

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Career Objective: Passionate and driven professional seeking a career in Research and Development to apply my expertise in developing innovative solutions, driving technological advancements, and contributing to cutting-edge projects that make a significant impact.

EDUCATION

Manipal Institute of Technology, Manipal, KA **2017-2021**
BTech, Instrumentation and Control | Minor: Sensor Technology

WORK EXPERIENCE

The Hi-tech Robotic Systemz, Gurugram, HR **Feb 2022 – Present**

Research Engineer *Oct 2022 – Present*

Multi-Floor Navigation

- Developed a custom ROS package that enabled a new feature of multi-floor navigation.
- Designed the package to dynamically switch maps based on the floor number trigger, integrated it into our SLAM (Simultaneous Localization and Mapping) module, and reset critical functionalities related to controls and navigation algorithms.
- Successfully released and deployed the package in production, enhancing our product's capabilities with multi-floor navigation as a new product feature.

Skills: ROS, C++.

Custom Obstacle Detection ROS Package

- Developed a custom obstacle detection package to enhance perception at designated drop locations using a 2D laser.
- Designed and implemented logic for defining a region of interest around drop locations, triggering new navigation goals.
- Successfully deployed the package, introducing a new feature that improved obstacle detection accuracy and enhanced navigation.

Skills: ROS, C++, PCL.

Development of Cobot for screwing application.

- Contributed to a proof-of-concept project for a UR10e robotic arm designed for automotive screwing applications.
- Responsible for motion planning, including inverse kinematics calculation and Cartesian path planning.
- Conducted "eye in hand" camera calibration for precise pose between end-effector and camera.
- Set up the apriltag_ros package for relative pose estimation.
- Built a real-time kernel to ensure instant servo control, that improved performance and reliability of system.

Skills: Python, ROS, MoveIt

Robotics Engineer – ROS Intern, Gurugram, HR *Feb 2022 – Oct 2022*

- Created a Unity-based digital twin for “Novus Carry 500” and “Novus Pallet Mover” mobile robots to facilitate product testing.
- Designed and developed URDF with a custom lifter attachment plugin for accurate simulation.
- Integrated 2-D lidar, 3-D lidar, and depth camera sensors for data visualization.
- Achieved a more efficient and cost-effective product development process, improving overall product performance.
- Streamlined product testing in a virtual environment, reducing time and costs.
- **Skills:** Unity, C#, ROS.

PERSONAL PROJECTS

ROS2-Based Gantry Car Painting Robot Development *May 2023 – Jul 2023*

- Collaborated in the development of a gantry-based car painting robot arm as part of the ROS2 Developer Learning program, under the mentorship of Lentin Joseph.
- Set up a Gazebo simulation environment to mimic real-world conditions.
- Utilized a depth camera for region of interest (ROI) detection and image segmentation.
- Concatenated point-clouds to reconstruct car as an Octomap for obstacle avoidance.
- Integrated the collected data for precise Cartesian path planning and painting actions.
- Accomplished a successful project that combined ROS2 and advanced image processing, enhancing painting efficiency.

Skills Learnt: ROS2, Gazebo, MoveIt2.

" **Vitarana Drone": Delivery Drone** *May 2020 – Mar 2021*

- Simulated autonomous parcel delivery using a drone within the Gazebo environment.
- Developed and implemented a PID algorithm for position and attitude control of the drone.
- Objectives included optimizing parcel delivery to specific locations on the map, aiming for maximum deliveries in minimal time.
- This project was a part of the MOOC initiative by IIT Bombay, contributing to my experience in robotics and autonomous systems.

Skills Learnt: ROS, Gazebo.

Skills & Interest

- Technical: C++, Python, Git, Docker, ROS1, PCL, OpenCV, Gazebo, MoveIt, Unity.
- Hardware: Jetson, 2D Lidar, 3D Lidar, Depth Camera.
- Languages: Fluent English, Hindi and Conversational Kannada.
- Interests: Cricket, Basketball.